Generative Artificial Intelligence and Entry-Level Skill Demand:

An Empirical Analysis of ChatGPT's Impact on the Labor Market

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1. Introduction

The launch of ChatGPT in November 2022 represented a significant development in the accessibility of Generative AI (GenAI) technology, reaching 100 million users within just two months. Recent research estimates that approximately 80% of the US workforce has at least 10% of their work tasks exposed to large language models (Eloundou et al., 2023). This widespread exposure broaches fundamental questions about how the labor market will adjust, particularly in entry-level hiring where employers must rely heavily on skill signaling to screen candidates that lack established performance records.

Entry-level labor markets present a unique context for studying GenAI's impact. Many new graduates signal productivity through demonstrated skills whether it is coding ability, writing samples, or analytical capabilities. However, if large language models can now perform many of these tasks, the informational value of traditional signals may be substantially diminished, creating an adverse selection problem where employers struggle to distinguish genuine ability from AI-assisted performance.

This study empirically examines whether and how employer demand for different skill types has shifted in entry-level positions following the economy's rapid GenAI adoption. I examine job posting data from 2021-2024, testing whether employers have adjusted skill requirements following ChatGPT's November 2022 launch. The central hypothesis is that employer demand for high AI-substitutable skills declined relative to demand for low AI-substitutable skills following ChatGPT’s release, with heterogeneous effects across occupation groups.

I employ a difference-in-differences regression on the Web Data Commons JobPosting Corpus, a dataset of structured postings extracted from the Common Crawl. I measure the

prevalence of different skills in entry-level job postings (bachelor's required, junior-level indicators) across occupation groups and years, classifying skills by AI-substitutability using existing exposure indices. This design allows for the comparison of high and low AI-substitutable skills before and after November 2022.

This research addresses critical policy needs as students are making significant education investments with incomplete information. Furthermore, universities require 2-4 years for curriculum updates, making early identification of skill demand shifts essential for proactive adaptation. My findings offer evidence-based guidance, for example, if demand for basic coding has declined while system design has increased, computer science departments can reallocate resources accordingly to properly prepare their students for a smoother transition into their professional careers.

My contribution features three key aspects. First, I provide a crucial skill-level decomposition of GenAI's impact, building on existing occupation-level research that cannot tell curriculum designers which specific competencies to emphasize. Second, I focus on entry-level positions where skill signaling is most critical. Third, I provide actionable evidence during a time in this phase of GenAI adoption where institutions can still proactively adapt.

2. Literature Review

2.1 Theoretical Framework: Acemoglu and Restrepo (2019)

Acemoglu and Restrepo (2019) developed the task-based theoretical framework structuring my analysis. Their model identifies why automation does not necessarily reduce overall employment despite substituting for labor in certain tasks through two opposing forces: displacement and reinstatement.

The displacement effect occurs when technology substitutes for labor in tasks that workers previously performed, directly reducing labor demand. The reinstatement effect operates through technological advances creating entirely new tasks (computers created demand for programmers) as well as productivity gains increasing output enough to raise labor demand even for partially automated tasks. Their empirical analysis of US labor markets from 1947-2017 shows manufacturing experienced strong displacement with weak reinstatement, while professional services demonstrated stronger reinstatement effects.

In the context of GenAI, displacement predicts that skills large language models substitute for, such as basic coding, report writing, and data entry, should decline in employer demand. Reinstatement can be seen through new tasks (prompt engineering, AI output evaluation) and complementary tasks increasing in value (system design, debugging, strategic thinking). My hypothesis tests whether displacement dominates at the aggregate level for entry-level skills, with Acemoglu and Restrepo providing the economic logic for why certain skills should move in opposite directions.

2.2 Methodological Foundation: Autor, Levy, and Murnane (2003)

Autor, Levy, and Murnane (2003) pioneered the empirical methodology for studying technology's impact at a granular level. Their central contribution was classifying tasks by susceptibility to computerization. More specifically, the idea that routine tasks "can be accomplished by machines following explicit programmed rules" while non-routine tasks cannot. The "routine task hypothesis" predicts computerization substitutes for routine cognitive tasks while complementing non-routine analytical tasks. Their empirical strategy combined task content measures with census employment data from 1960-1998, finding employment in routine

cognitive occupations declined 3.2 percentage points and non-routine analytical occupations increased 7.1 percentage points.

I extend their methodology along three dimensions. First, I shift from task content to skill demand in job postings, the policy-relevant outcome for education. Second, GenAI represents a fundamental shift from computerization. Large language models can draft original prose and generate code for novel problems, tasks Autor et al. classified as non-routine. I develop an updated classification based on AI-substitutability rather than routine-ness. Third, I exploit ChatGPT's sharp November 2022 launch to act as an immediate transformation rather than gradual technology adoption.

2.3 AI Exposure Measurement: Felten, Raj, and Seamans (2021)

Felten, Raj, and Seamans (2021) developed the first systematic methodology for associating AI technological capabilities to occupational requirements. Their innovation was decomposing AI progress into specific capability domains (image recognition, language understanding, speech processing) and mapping these to abilities required in different occupations using O*NET data. They constructed AI Occupational Exposure indices by evaluating whether advances in each AI domain complement or substitute for abilities listed in O*NET, creating occupation-level exposure scores that vary by AI technology type.

Their methodological contribution provides a structured framework for translating technological capabilities into labor market metrics, revealing that AI exposure is multifaceted. For my research, Felten et al. provide the conceptual template for constructing AI-substitutability classifications. I adapt their ability-level framework to skill-level analysis, assessing which skills in job postings are most and least threatened by large language models in particular.

2.4 Recent Empirical Evidence: Eloundou et al. (2023)

Eloundou et al. (2023) provide the most comprehensive occupation-level measurement of LLM exposure. They employed both human experts and GPT-4 to evaluate whether LLM access would cut task completion time by 50% for all work activities detailed in O*NET.

The paper's key findings reveal widespread exposure: approximately 80% of workers have at least 10% of tasks exposed to LLMs, with 19% having at least 50% exposed. High-exposure occupations include writers (69%) and mathematicians (67%), while low-exposure occupations include agricultural equipment operators (1%) and construction trades (4%). This pattern inverts historical automation, which primarily affected manufacturing and routine manual work. Cognitive, language-intensive, white-collar occupations now face high exposure, and higher wages generally correlate with higher exposure. Workers with bachelor's degrees face particularly high exposure. Industry analysis further reinforced these patterns: data processing (84% of workers highly exposed), publishing (79%), and legal services (75%) face the greatest exposure.

The critical gap emerges from this occupation-level focus. While Eloundou et al. report that software developers are 65% exposed, this information is simply insufficient for education policy. Computer Science programs aim to develop specific competencies, such as algorithms, system design, debugging, and testing, with each area potentially being impacted differently by GenAI. Knowing the occupation is "65% exposed" does not actually indicate which competencies to prioritize.

My research bridges this gap through skill-level analysis. If demand for system architecture is increasing while demand for writing CRUD operations is declining, this directly informs curriculum decisions. The skill-level decomposition translates Eloundou et al.'s macro-level exposure into actionable micro-level guidance. Furthermore, my focus on

entry-level positions addresses a unique population where employers are forced to rely on interpreting skill signals rather than professional history when hiring.

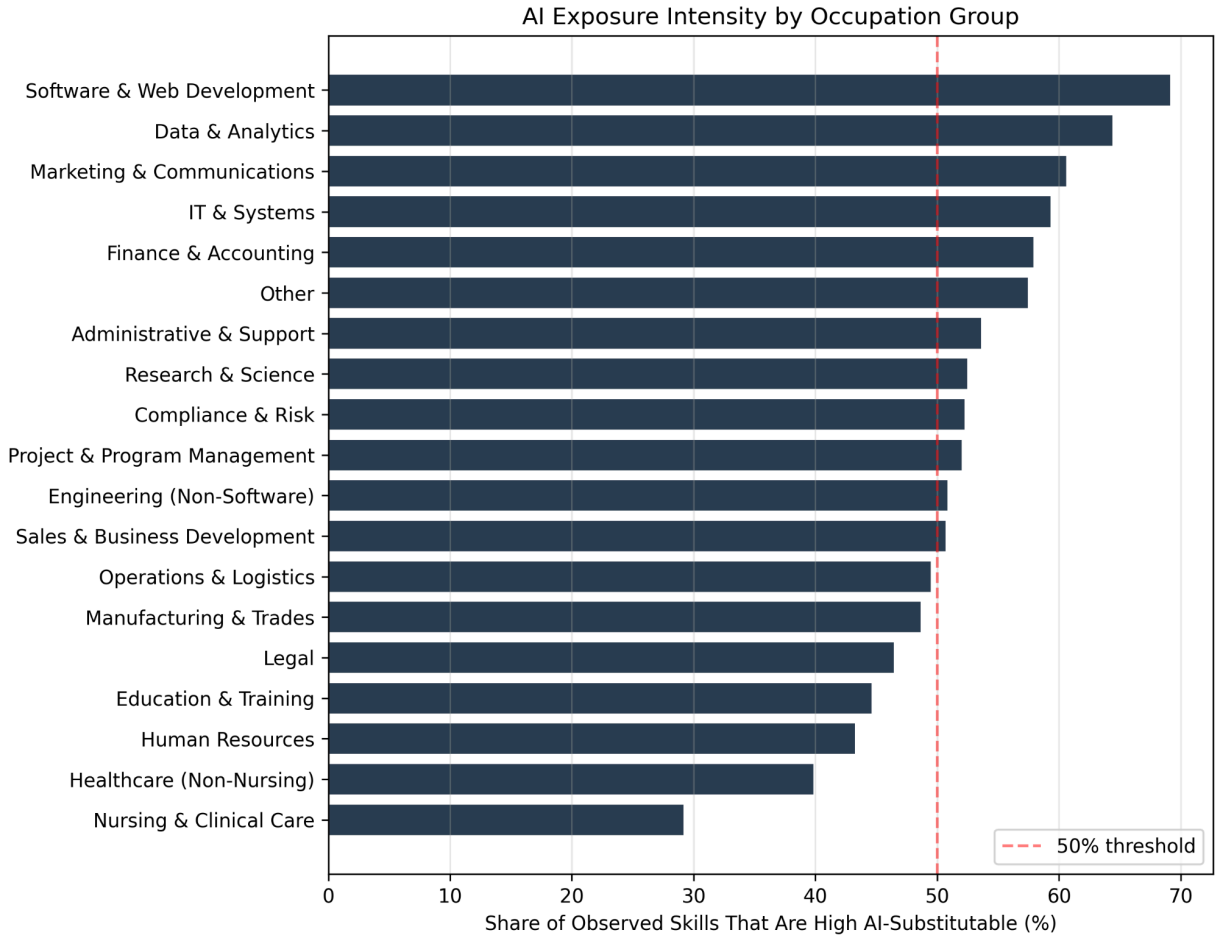
3. Data

This study uses the Web Data Commons (WDC) JobPosting Corpus, a dataset of structured job postings extracted from the Common Crawl. The WDC project parses websites for schema.org/JobPosting markup, which is structured metadata that employers and job boards embed in their HTML, yielding real postings from platforms such as Indeed, Glassdoor, and ZipRecruiter rather than self-reported survey data. I construct four annual cross-sections (2021–2024) totaling 18,927 postings after applying three sequential filters: U.S. postings only, full-time positions, and entry-level roles requiring a bachelor's degree with at least one junior-level indicator. From each filtered year, a stratified reservoir sample of approximately 4,750 postings is drawn. Six text fields per posting (description, qualifications, skills, experience requirements, education requirements, and responsibilities) are used for skill extraction.

An automated pipeline, powered by Anthropic's Claude Opus 4.6, identifies mentions of 279 standardized skill labels across each posting's six text fields. Each posting is also classified into one of 19 occupation groups using a priority-ordered regex classifier applied to job titles, ensuring consistent labeling across years (*Table A1*). The 279 skills are split into a treatment group of 162 high AI-substitutable skills, meaning tasks an LLM can perform at entry level (coding, data analysis, content writing), and a control group of 117 low AI-substitutable skills requiring physical presence, human interaction, or licensed judgment (patient care, welding, leadership), informed by Eloundou et al. (2023). The treatment shock is ChatGPT's November 2022 release (pre: 2021–2022; post: 2023–2024). Posting-level data is aggregated into a skill-occupation-year panel (279 × 19 × 4 = 21,204 observations) by computing each skill's

frequency within each occupation-year cell. Occupation is used as the cross-sectional control because skills cluster more tightly by occupation than by industry. For example, a Junior Financial Analyst and a Marketing Coordinator in the same industry require fundamentally different competencies, and *Figure 1* confirms that AI exposure intensity ranges from below 30% in Nursing to a majority in Software & Web Development.

Figure 1 - AI Exposure by Occupation Group using Skill-level Granularity



These data capture what employers actually ask for in job postings rather than what workers self-report or what occupational taxonomies assume, avoiding the biases that affect survey instruments, and the consistent 279-skill taxonomy ensures the same competency is measured identically across all four years. However, limitations include a selection bias toward

larger employers that implement schema.org markup, a bounded taxonomy that may miss emerging skills like "prompt engineering" (potentially understating the reinstatement effect of Acemoglu and Restrepo, 2019), sample compositional shifts as Software & Web Development postings declined from 423 to 128, and annual granularity limiting the precision of the treatment timing. Most prior studies operate at the occupation or task level: Eloundou et al. (2023) and Felten, Raj, and Seamans (2021) aggregate across all workers within an occupation and cannot distinguish between two "Junior Financial Analyst" postings that differ in whether they require Python or only Excel. Task-level analyses using O*NET experience a related problem, as a task like "analyze financial data" remains static regardless of whether employers now expect it to be completed with the help of an LLM. Skill-level analysis on job postings observes employer demand directly and in real time, and the four-year panel enables a difference-in-differences identification strategy that prior cross-sectional analyses cannot support.

Table 1 reports descriptive statistics. Mean skill frequency is 2.39% (SD = 6.51 pp), with high AI-substitutable skills appearing at roughly half the rate of low AI-substitutable skills (1.75% vs. 3.28%), a compositional artifact of the treatment group containing specialized technical skills while the control group includes broadly demanded interpersonal competencies. *Figure 2* plots the group means over time and reveals two stylized facts. Both groups move roughly in parallel from 2021 to 2022, supporting the parallel trends assumption. After ChatGPT's release, the groups diverge: control skills dip in 2023 before recovering above their 2021 level by 2024 (3.37%), while treatment skills rise slightly in 2023 and then fall to 1.67% in 2024. One might expect demand for AI-substitutable skills to drop sharply and immediately, but the non-monotonic trajectory suggests a diffusion lag. Whether this aggregate pattern holds

across all occupations or masks significant heterogeneity is the question the per-occupation DiD regressions in the following empirical analysis will address.

Figure 2 - Annual Mean Skill Frequency of Treatment and Control Groups

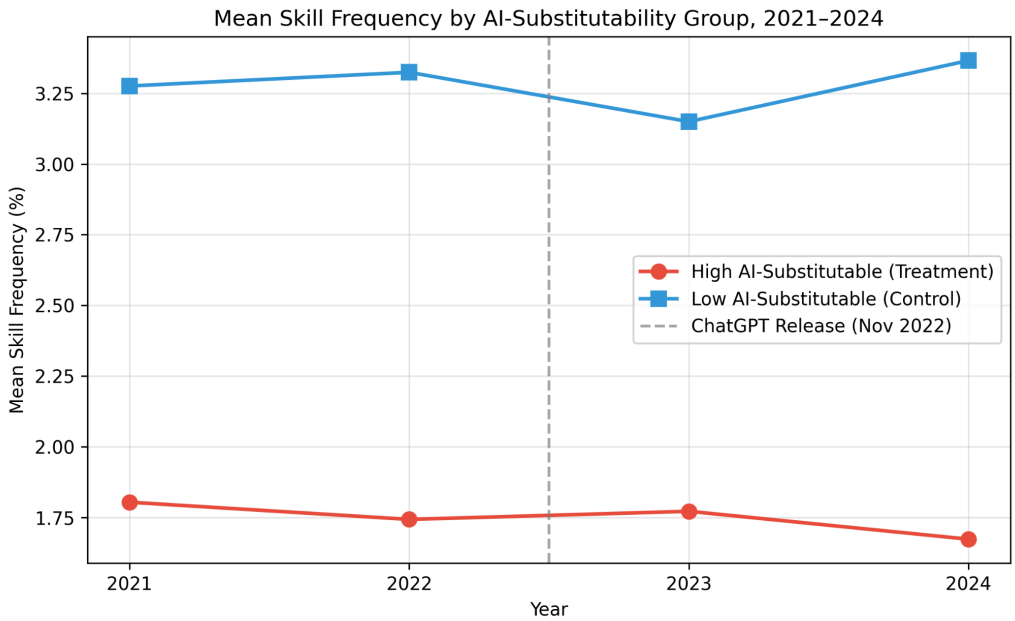


Table 1 - Descriptive Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Skill frequency (%, all)	21,204	2.39	6.51	0.00	88.28
High AI-substitutable	12,312	1.75	5.41	0.00	88.28
Low AI-substitutable	8,892	3.28	7.70	0.00	76.79
Treated (=1 if high AI-sub)	21,204	0.58	0.49	0.00	1.00
Post (=1 if year >= 2023)	21,204	0.50	0.50	0.00	1.00

4. Econometric Model

4.1 Hypothesis

The displacement effect identified by Acemoglu and Restrepo (2019) predicts that when technology substitutes for labor in particular tasks, employer demand for the skills associated with those tasks should decline. In the context of Generative AI, large language models can now

perform entry-level coding, data analysis, and content writing at a meaningful quality. These are tasks that Eloundou et al. (2023) estimate account for a substantial share of white-collar entry-level work. This study hypothesizes that following ChatGPT's November 2022 release, employer demand for high AI-substitutable skills declined relative to demand for low AI-substitutable skills in entry-level job postings, with heterogeneous effects across occupation groups.

This hypothesis extends the existing literature in a specific and important manner. Eloundou et al. (2023) and Felten, Raj, and Seamans (2021) measure AI exposure at the occupation level, for example, estimating that software developers face 65% GPT exposure, but neither study tests whether employer demand for specific skills actually shifted following the shock. Autor, Levy, and Murnane (2003) established the empirical methodology for studying technology's impact at a granular level, but their routine task hypothesis was developed in the context of computerization rather than Generative AI, which can perform non-routine cognitive tasks that prior automation could not. This appears to leave a gap in the literature: no existing study uses a difference-in-differences design applied to entry-level job posting data to identify whether ChatGPT's release caused a measurable shift in skill-level demand. This analysis will make that contribution.

4.2 Econometric Model

To test this hypothesis, I estimate the following difference-in-differences specification:

$$skill_pct_{iot} = \beta_0 + \beta_1 \, Treated_{io} + \beta_3 \, (Treated_{io} \times Post_t) + \alpha_o + \lambda_t + \varepsilon_{iot}$$

The dependent variable, *skill_pct_iot*, is the frequency at which skill *i* appears in occupation *o* job postings during year *t*, expressed as a percentage. The panel contains 279 skills x 19 occupation groups x 4 years, yielding 21,204 observations. $Treated_{io}$ is an indicator equal to 1 for

the 162 high AI-substitutable skills and 0 for the 117 low AI-substitutable control skills, informed by the GPT exposure framework described in Eloundou et al. (2023). The coefficient of interest is β_3 on the interaction term $Treated_{io} \times Post_t$. A negative and statistically significant β_3 would indicate that AI-substitutable skill demand fell relative to the control group after ChatGPT's release, supporting the hypothesis.

A standalone $Post_t$ term is excluded from the specification because the year fixed effects λ_t already absorb all common time variation in a more flexible manner, assigning a separate intercept to each year with 2021 as the reference. This is preferable to a single *Post* dummy, which would assume the only common time variation exists within the discrete pre/post jump. Given that the sample spans post-COVID labor market recovery, the 2023 tech sector contraction, and the continued GenAI diffusion through 2024, year fixed effects better isolate the differential treatment effect from these distinct annual shocks. No variable transformations are applied as skill frequency is already a bounded proportion and the hypothesis concerns percentage point shifts.

Why Not OLS

A simple OLS regression of skill frequency on a post-ChatGPT dummy would be biased due to omitted variable bias. Occupations systematically differ in which skills they demand independent of AI substitutability. For instance, Software & Web Development postings inherently list more coding skills than Nursing postings because of the nature of the work, not because of ChatGPT. This compositional difference is visible in *Table 1*: high AI-substitutable skills appear at 1.75% on average compared to 3.28% for control skills. OLS cannot separate these level differences from a true post-ChatGPT shift, whereas the difference-in-differences design identifies β_3 from changes within occupation groups over time.

Identification Strategy

The causal interpretation of β_3 is based on three conditions. First, the treatment is exogenous as ChatGPT's November 2022 release was an external shock whose timing was not endogenous to any occupation or skill group. Second, the parallel trends assumption must hold, and *Figure 2* provides supporting evidence; both groups track closely from 2021 to 2022 before diverging after release. Third, there must be no concurrent confounders that differentially affected high AI-substitutable skills but not low AI-substitutable skills. Occupation fixed effects α_o absorb time-invariant occupation-level confounders and year fixed effects λ_t absorb macro shocks common to all skills. Together these conditions support a causal interpretation of β_3: the divergence in skill frequency between treatment and control groups following ChatGPT's release reflects the causal effect of GenAI substitutability on employer demand, rather than pre-existing differences or coincidental time trends.

Econometric Issues

The primary concern is omitted variable bias from unobserved skill-level trends predating 2023. If treated skills were already declining pre-2023 for reasons unrelated to Generative AI, β_3 would be biased negative, meaning any significant negative estimate is conservative. On the other hand, if they were already rising, it would bias toward zero. The pre-period parallel trends in *Figure 2* partially address this, and the event study in Section 5 formally tests the assumption. Standard errors are clustered at the occupation level to account for within-occupation error correlation across skills and years, and are robust to heteroscedasticity. Since the number of occupation groups (19) falls below the conventional threshold of 30, the clustered standard errors should be interpreted with some caution. Finally, multicollinearity is not a concern as VIF for *Treated* is 4.24 and for *Treated x Post* is 4.50, with the moderate elevation expected given the

interaction term is constructed from *Treated* and does not indicate a problematic degree of collinearity.

Robustness Checks

Three robustness checks are performed: (1) an event study specification with year-specific interaction terms to test parallel trends and trace the dynamic pattern of treatment effects; (2) per-occupation DiD regressions to test heterogeneity across the 19 occupation groups; and (3) redefining Post to equal 1 starting in 2024 only, to test the diffusion lag interpretation suggested by the non-monotonic 2023 trajectory illustrated in *Figure 2*.

5. Results

5.1 Main Results

Table 2 presents the DiD estimates across five specifications in order of increasing complexity. The coefficient of interest, β_3 on the *Treated* × *Post* interaction, is negative across all specifications but statistically insignificant in the pooled 2023-2024 post period. In the main two-way fixed effects specification (Column 4), $\beta_3 = -0.009$ ($p = 0.940$), indicating no detectable aggregate effect of ChatGPT's release on AI-substitutable skill demand when the full post period is considered. The baseline level difference between treatment and control groups, captured by β_1, is -1.528 ($p < 0.001$) across all specifications, confirming the compositional gap documented in *Table 1* but reflecting skill-type differences rather than a treatment effect. The R^2 of 0.023 is low across all specifications, which is expected as most variation in skill frequency is idiosyncratic to individual skills, and the goal of the model is causal identification rather than prediction.

Table 2 - DiD Estimates: Effect of ChatGPT on AI-Substitutable Skill Demand

	Base	**Occ FE**	**Year FE**	**Two-Way FE**	**Post = 2024**
β_1	-1.507*** (0.252)	-1.507*** (0.252)	-1.528*** (0.259)	-1.528*** (0.259)	-1.478*** (0.284)
β_3	-0.051 (0.081)	-0.051 (0.081)	-0.009 (0.119)	-0.009 (0.119)	**-0.216*** (0.114)
R^2	0.013	0.023	0.013	0.023	0.023
Occ FE	No	Yes	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes
N	21,204	21,204	21,204	21,204	21,204

*Clustered Standard Errors in Parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$*

5.2 Event Study

To examine the dynamic pattern of treatment effects and formally test the parallel trends assumption, I estimate an event study specification replacing the single Post indicator with year-specific interaction terms, using 2021 as the reference year:

$$skill_pct_{iot} = \beta_0 + \beta_1 \, Treated_{io}$$

$$+ \beta_{22} \, (Treated_{io} \times 2022) + \beta_{23} \, (Treated_{io} \times 2023) + \beta_{24} \, (Treated_{io} \times 2024)$$

$$+ \alpha_o + \lambda_t + \varepsilon_{iot}$$

The results are reported in *Table 3*. The 2022 pre-treatment coefficient is −0.109 (p = 0.312), providing formal statistical support for the parallel trends assumption by indicating there is no evidence of differential pre-treatment movement between high and low AI-substitutable skills prior to ChatGPT's release. The 2023 coefficient is +0.094 (p = 0.647), insignificant and slightly positive, indicating no immediate demand response in the year following ChatGPT's launch. The 2024 coefficient is −0.221 (p = 0.048), negative and statistically significant at the 5% level. This

is the study's strongest result: AI-substitutable skill demand declined by 0.221 percentage points relative to control skills in 2024, the first year in which a significant effect is detectable.

Table 3 - Event Study Estimates: Dynamic Treatment Effects by Year

	Base	**Occ FE**	**Year FE**	**Two-Way FE**
β_1	-1.476*** (0.255)	-1.476*** (0.255)	-1.473*** (0.252)	-1.473*** (0.252)
β_{22}	-0.061 (0.071)	-0.061 (0.071)	-0.109 (0.108)	-0.109 (0.108)
β_{23}	-0.032 (0.135)	-0.032 (0.135)	+0.094 (0.205)	+0.094 (0.205)
β_{24}	-0.131** (0.065)	-0.131** (0.065)	-0.221** (0.112)	-0.221** (0.112)
R^2	0.014	0.023	0.014	0.023
Occ FE	No	Yes	No	Yes
Year FE	No	No	Yes	Yes
N	21,204	21,204	21,204	21,204

*Clustered Standard Errors in Parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$*

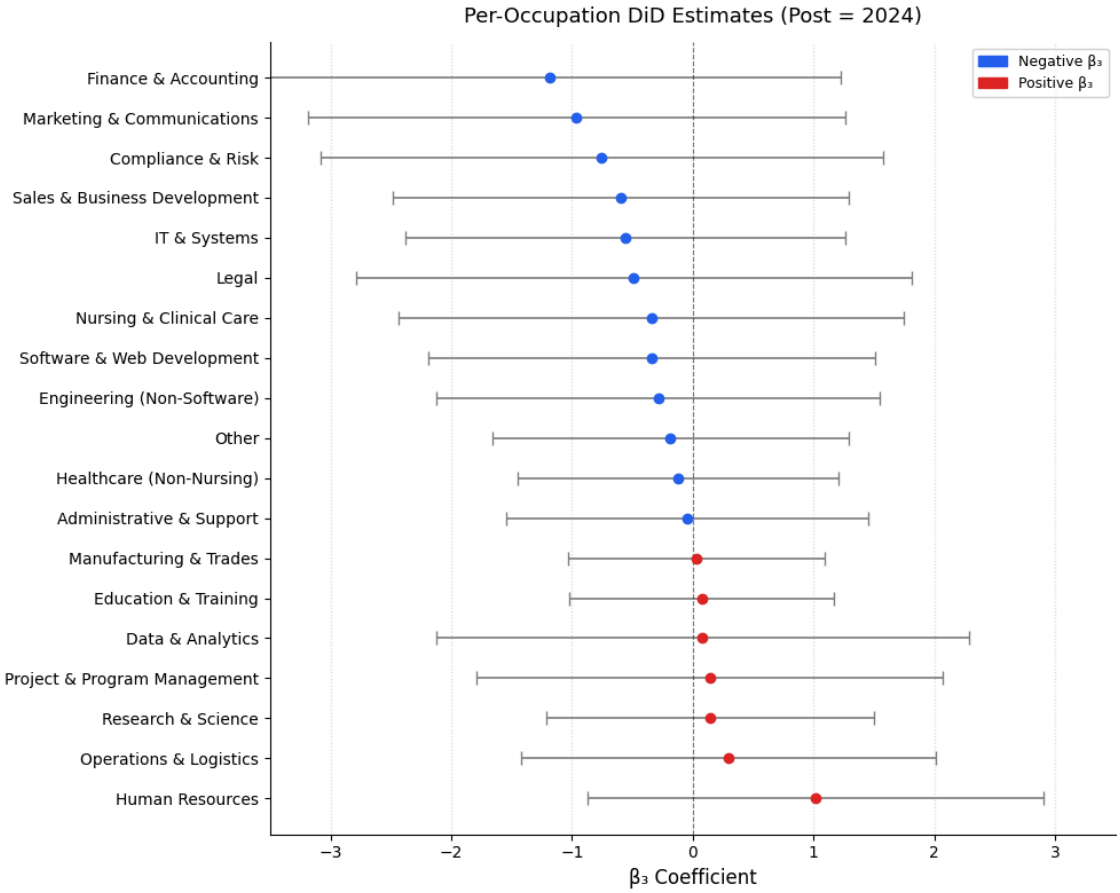
This pattern is consistent with a diffusion lag. Rather than immediately revising skill requirements upon ChatGPT's release, employers appear to have adjusted their hiring criteria only after generative AI tools became sufficiently integrated into enterprise workflows. The near-zero 2023 coefficient suggests that early 2023 postings largely predated meaningful enterprise adoption, while the significant 2024 decline reflects a more systematic reassessment of entry-level skill requirements as these tools matured.

5.3 Robustness Checks

Three robustness checks are reported alongside the main specification in *Table 2*. First, redefining Post as 2024 only in the standard DiD framework yields $\beta_3 = -0.216$ ($p = 0.059$),

marginally significant at the 10% level and consistent with the event study finding that the effect concentrates in 2024. Second, the base specification without fixed effects and the occupation-only fixed effects specification both yield $\beta_3 = -0.051$ ($p = 0.526$), confirming that the near-zero pooled result is not an artifact of the fixed effects structure. Third, as reported in *Figure 3* and *Table A2*, per-occupation DiD regressions with Post = 2024 show that 13 of 19 occupation groups produce negative β_3 estimates, with the largest effects in Finance & Accounting (−1.182), Marketing & Communications (−0.962), and Compliance & Risk (−0.756), all three of which were high AI-exposure occupations per *Figure 1*. However, no individual occupation produces a statistically significant result, reflecting limited statistical power at the occupation level given the small number of observations per cell.

Figure 3 - Forest Plot of Per-Occupation DiD Estimates, Post = 2024



5.4 Comparison to Prior Literature

The null aggregate result and delayed 2024 effect are not inconsistent with prior literature. Eloundou et al. (2023) and Felten, Raj, and Seamans (2021) measure AI exposure at the occupation level but do not test whether employer demand actually shifted following a specific technological shock. High measured exposure does not necessarily imply an immediate labor market response. Autor, Levy, and Murnane (2003) found that computerization's effects on routine task employment accumulated gradually over decades rather than appearing immediately, and Acemoglu and Restrepo (2019) similarly document that displacement effects take time to materialize in observable labor market outcomes. The concentration of effects in 2024 rather than 2023 is consistent with this historical pattern of gradual adjustment following technological shocks.

5.5 Policy Implications

The findings carry several implications for education and workforce policy. The absence of a significant aggregate effect through 2023 suggests that universities and curriculum designers should not make immediate overarching changes to AI-substitutable skill offerings. The demand shift, if real, is gradual rather than abrupt. However, the significant 2024 result and the directional occupation-level evidence suggest that monitoring entry-level job posting data beyond 2024 is warranted, as the adjustment may be accelerating as enterprise GenAI adoption matures. The concentration of negative effects in Finance & Accounting, Marketing & Communications, and Compliance & Risk suggests that targeted curriculum reviews in those fields are more justified than blanket reductions across all programs. Finally, the modest effect in Software & Web Development despite its high AI-exposure classification is consistent with Acemoglu and Restrepo's reinstatement effect. Complementary technical skills such as system

design and debugging may be increasing in value even as more routine coding tasks become automatable, suggesting that curriculum designers in computer science should prioritize higher-order competencies over routine implementation skills.

6. Conclusion

This paper examined whether ChatGPT's November 2022 release shifted employer demand for AI-substitutable skills in entry-level job postings. Using a skill-occupation-year panel of 21,204 observations constructed from the Web Data Commons JobPosting Corpus, I estimated a difference-in-differences specification comparing 162 high AI-substitutable skills to 117 low AI-substitutable skills across 19 occupation groups from 2021 to 2024.

The pooled DiD estimate is a precisely estimated null (β_3 = -0.009, p = 0.940), but the event study reveals that this aggregate result masks a meaningful temporal pattern. The 2022 pre-treatment coefficient is statistically indistinguishable from zero, supporting the parallel trends assumption. The 2023 coefficient is also near zero, indicating no immediate employer response. The 2024 coefficient is -0.221 and statistically significant at the 5% level, consistent with a diffusion lag in which employers revised skill requirements only after GenAI tools became sufficiently integrated into enterprise workflows. Per-occupation regressions reinforce this interpretation directionally, with 13 of 19 occupations showing negative point estimates and the largest declines concentrated in Finance & Accounting, Marketing & Communications, and Compliance & Risk.

Altogether, the evidence supports a qualified version of the displacement hypothesis. GenAI's effect on entry-level skill demand is real but delayed, emerging by 2024 rather than immediately following the November 2022 shock. This pattern is consistent with the historical record of gradual adjustment following technological shocks documented by Autor, Levy, and

Murnane (2003) and Acemoglu and Restrepo (2019). It also suggests that the occupation-level exposure scores of Eloundou et al. (2023) and Felten, Raj, and Seamans (2021) are better interpreted as forecasts of eventual demand shifts rather than indicators of immediate labor market reallocation.

The study has several limitations worth noting. The 19 occupation clusters fall below the conventional threshold of 30 for clustered standard errors, so all inference should be interpreted with appropriate caution. Annual data granularity limits precision around the treatment timing, as postings scraped in early 2023 likely predate meaningful enterprise GenAI adoption. The 279-skill taxonomy is bounded and may miss emerging competencies such as prompt engineering, potentially understating the reinstatement effect. Sample composition shifts, particularly the decline in Software & Web Development postings from 423 to 128, affect the precision of smaller-cell per-occupation estimates.

Future research should extend this analysis in three directions. First, as post-2024 data becomes available, the delayed effect identified here should be tracked to determine whether it is accelerating, stabilizing, or reversing. Second, the skill taxonomy should be expanded to include emerging GenAI-complementary competencies, allowing direct tests of the reinstatement effect that the current taxonomy understates. Third, richer occupational and firm-level heterogeneity could shed light on which employers adjust fastest and why, which is the question most directly relevant to curriculum designers and workforce policymakers. The core finding of this paper, that the labor market effects of GenAI are real but diffuse slowly, implies that the window for proactive curriculum adaptation remains open but is narrowing.

Data and Code Availability

Data and code are available at https://github.com/arinjay-singh/genai-entrylevel-skilldemand.

Bibliography

Acemoglu, Daron, and Pascual Restrepo. 2019. "Automation and New Tasks: How Technology Displaces and Reinstates Labor." Journal of Economic Perspectives, 33(2): 3-30.

Autor, David H., Frank Levy, and Richard J. Murnane. 2003. "The Skill Content of Recent Technological Change: An Empirical Exploration." The Quarterly Journal of Economics, 118(4): 1279-1333.

Eloundou, Tyna, Sam Manning, Pamela Mishkin, and Daniel Rock. 2023. "GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models." Science, 381(6659): eadi9837.

Felten, Edward W., Manav Raj, and Robert Seamans. 2021. "A Method to Link Advances in Artificial Intelligence to Occupational Abilities." AEA Papers and Proceedings, 111: 54-57.

Appendix A. Supplementary Tables

Table A1 - Sample Composition by Occupation Group and Year

Occupation	2021	2022	2023	2024	Total
Administrative & Support	502	594	611	723	2430
Compliance & Risk	102	128	84	98	412
Data & Analytics	282	216	198	198	894
Education & Training	201	430	328	274	1233
Engineering (Non-Software)	160	151	149	180	640
Finance & Accounting	494	493	896	467	2350
Healthcare (Non-Nursing)	166	125	207	265	763
Human Resources	62	60	38	55	215
IT & Systems	151	186	134	145	616
Legal	71	68	52	57	248

Manufacturing & Trades	404	491	296	402	1593
Marketing & Communications	202	171	162	148	683
Nursing & Clinical Care	168	113	130	199	610
Operations & Logistics	201	252	147	171	771
Other	600	567	838	816	2821
Project & Program Management	**114**	**106**	**109**	**108**	**437**
Research & Science	**119**	**60**	**80**	**111**	**370**
Sales & Business Development	**328**	**174**	**154**	**205**	**861**
Software & Web Development	**423**	**292**	**137**	**128**	**980**
Total	**4750**	**4677**	**4750**	**4750**	**18927**

Table A2 - Per-Occupation DiD Estimates: Effect of ChatGPT on AI-Substitutable Skill Demand, Post = 2024

Occupation	β_3	SE	P-value	N	R^2
Finance & Accounting	-1.182	1.226	0.335	1116	0.003
Marketing & Communications	-0.962	1.136	0.397	1116	0.008
Compliance & Risk	-0.756	1.187	0.524	1116	0.006
Sales & Business Development	-0.601	0.963	0.532	1116	0.048
IT & Systems	-0.556	0.929	0.549	1116	0.006
Legal	-0.488	1.172	0.677	1116	0.031
Nursing & Clinical Care	-0.345	1.069	0.747	1116	0.049
Software & Web Development	-0.342	0.945	0.718	1116	0.013
Engineering (Non-Software)	-0.286	0.936	0.760	1116	0.052
Other	-0.185	0.752	0.805	1116	0.029

Healthcare (Non-Nursing)	-0.122	0.678	0.857	1116	0.058
Administrative & Support	-0.049	0.764	0.949	1116	0.038
Manufacturing & Trades	0.029	0.544	0.958	1116	0.042
Education & Training	0.073	0.558	0.895	1116	0.031
Data & Analytics	0.081	1.124	0.943	1116	0.000
Project & Program Management	0.139	0.985	0.888	1116	0.023
Research & Science	0.146	0.692	0.833	1116	0.020
Operations & Logistics	0.291	0.876	0.740	1116	0.036
Human Resources	1.016	0.962	0.291	1116	0.029